Exercise 8: Multivariate Optimization 3 (SGD)

Optimization in Machine Learning

> **Reproducibility**
>
> Produced with Python 3.10 and R 4.4.1. To reproduce, install the pinned package versions below.
>
> **Python**
>
> ```
> pip install numpy==2.0.2 matplotlib==3.10.8
> ```
>
> **R**
>
> ```
> install.packages("remotes")
> remotes::install_version("ggplot2", "3.5.1")
> remotes::install_version("gridExtra", "2.3")
> ```

Consider ordinary linear least squares (without intercept):

$$\min_{\theta} \mathbb{E}_{\mathbf{x},y}\left[(\theta^\top \mathbf{x} - y)^2\right]$$

with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\mathbf{x}})$ and $y \mid \mathbf{x} \sim \mathcal{N}({\theta^*}^\top \mathbf{x}, \sigma^2)$. Throughout the coding parts we use the univariate setting $\Sigma_{\mathbf{x}} = 1/4$, $\sigma = 1/10$, $\theta^* = 1/2$. Each coding cell draws fresh samples from this generative model (fresh mini-batches in (c); streaming single samples in (e)).

> **Note**
>
> **R vs. Python parity.** R's `set.seed(123)` and Python's `np.random.seed(123)` use *different* RNGs. The data above (and consequently the SGD trajectories below) differ between languages. Both implementations are correct; only the underlying data differs. Compare *algorithmic behavior* (cloud shape, convergence pattern) rather than expecting

numerically identical θ values.

(a) Unbiasedness of the SGD gradient

Show that

$$\mathbb{E}_{\mathbf{x},y}\left[\nabla_\theta\left[(\theta^\top\mathbf{x}-y)^2\right]\right] = \nabla_\theta\mathbb{E}_{\mathbf{x},y}\left[(\theta^\top\mathbf{x}-y)^2\right].$$

(b) Implication for SGD

Interpret (a) in terms of stochastic gradient descent.

(c) Gradient variance simulation

In the univariate setting $\Sigma_{\mathbf{x}} = 1/4$, $\sigma = 1/10$, $\theta^* = 1/2$, with data of size 10,000:

Plot the **confusion** (variance of the SGD gradient around the true full-batch gradient) for $\theta \in \{0, 0.05, 0.10, \dots, 1.00\}$. For each θ draw 200 independent batches and plot the per-batch confusion. Repeat for batch sizes $m = 100$ and $m = 1{,}000$.

(d) Reading the simulation

What do you observe in (c)?

(e) SGD vs population GD

On the same setup, run SGD with **batch size 1**, step size $\alpha = 0.3$, starting at $\theta = 0$, for 20 iterations. Repeat the SGD run 200 times and overlay all trajectories. Compare with **population GD** (same step size, same start) using the analytical gradient $g(\theta) = 2\sigma_x^2(\theta - \theta^*)$.